CSE 4820 / CSE 5819 Fall 2025-Section 001
Instructor: Jinbo Bi
Introduction to Machine Learning

Assignment 8 (100 pts)
Due: Nov 23rd (Sunday) mid-night
Neural Networks, Multi-layer Perceptron, RNN, CNN

Name: ____________________ NetID: _____________________

Part 1: (ChatGPT Self-Learning) 20pts

This semester we are trying to use some generative AI to help learn ML concepts. Keep in mind that generative AI does not always search from authoritative websites and does not always answer the questions correctly. It is your responsibility to judge and learn from multiple sources based on our in-class discussion. You can start from the following prompts and create subsequent questions that attempt to understand the ML concepts. For each question you have asked chatGPT, just copy the first three lines of its answer into your answer sheet (so to save the space but show that you have studied). Please include this part in the end of your HW after you answer HW part 2.

1. What are neural networks?
2. What is the difference between neural networks and artificial neural networks?
3. How many kinds of artificial neural networks?
4. What is a neuron?
5. What does an activation function do?
6. What is a layer in neural networks?
7. What is a loss function when training a neural network?
8. Why single layer perceptron cannot approximate XOR logic function?
9. Why multi-layer perceptron can approximate many functions (universal appximation theorem)?
10. What is a recurrent neural network?
11. What is a convolution neural network?
12. What does it mean backpropagation?
13. What does backpropagation do?
14. How are the weights of a neural network determined or optimized?
15. What is the problem of gradient vanishing and gradient explostion?
16. Why might ReLU be preferred in very deep networks?
17. What is the difference between a CNN, RNN, and Transformer?
18. What is a Transformer?

One of the problems here will be tested in the final exam. Please do study.

Part 2: Answer the Following Problems (80pts).

This part will be graded based on correctness, accuracy and clarify. Please prepare your answers in a pdf file to submit through HuskyCT assignment portal and clearly label your file with part 2 of assignment number. If you decide to use handwriting, make sure your handwriting is readable; or otherwise TAs have all rights to give 0 pt for answers that they cannot read. Please provide your calculation process, based on which you may get partial scores even if your answer is not correct.
Please try not to use ChatGPT (or other generative AI) to answer this part. If you rely too much on generative AI for this part, you may not learn enough and be well prepared for exams.

[**Recurrent neural networks**] [25pts]

You are given a simple recurrent neural network designed to process short character sequences. The network takes as input a sequence of characters from the alphabet $\{A, B\}$, encoded as one-hot vectors, and predicts the next character in the sequence.

The RNN has the following structure:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b_h)$$

$$\hat{y}_t = \text{softmax}(W_y h_t + b_y)$$

where

- $x_t \in \mathbb{R}^2$ is the one-hot input at time t,
- $h_t \in \mathbb{R}^3$ is the hidden state,
- $\hat{y}_t \in \mathbb{R}^2$ is the predicted probability distribution for the next character.

Assume $h_0 = 0$.

1. Can you compute the forward pass? (16pts)

You are given the following parameter matrices:

$$W_x = \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ -1 & 1 \end{bmatrix}, \quad W_h = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix}$$

$$W_y = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & -1 \end{bmatrix}, \quad b_h = 0, \; b_y = 0$$

You receive the 3-character input sequence:

A, B, A

(a) Convert each character to a one-hot vector.

(b) Compute h_1, h_2, and h_3.

(c) Compute the predicted outputs $\hat{y}_1$, $\hat{y}_2$, and $\hat{y}_3$.

(d) Which character does the model predict after the final timestep?

2. Can you perform gradient reasoning? (14 pts)

For one timestep t, derive the gradient of the loss L_t with respect to the hidden state h_t.

Show that:

$$\frac{\partial L}{\partial h_t} = W_y^\top(\hat{y}_t - y_t) + W_h^\top\left(\frac{\partial L}{\partial h_{t+1}} \odot (1 - h_t^2)\right)$$

Explain in your own words how this equation illustrates the **vanishing/exploding gradient** problem.

[Convolution Neural Networks]
[15 pts]

1(a). Suppose you have a CNN layer with d_0 input channels and d_1 output channels, with a filter of size $k \times k$ and stride s in both directions. When you apply this layer to an input I of size $n \times n \times d_0$, with padding p in both directions, what is the dimension of the output feature map O?

The network looks as follows (where *in* is the input, *conv* is the CNN operator mentioned above, and *out* is the output

$$\boxed{in} \xrightarrow{\quad conv \quad} \boxed{out}$$

[15pts]

1(b). Suppose you are given a 3×3 matrix from one patch of an image I and a 2×2 convolutional kernel K. In the image matrix, each entry represents the grayscale colors of a corresponding pixel.

$$I = \begin{bmatrix} 5 & 1 & 1 \\ 3 & 0 & 2 \\ 4 & 4 & 0 \end{bmatrix}$$

and

$$K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

What is the output of applying the filter to the input with stride 1 without padding?

Part 3 [**Programming**] (20 pts) In earlier HWs, you have used Seaborn Iris dataset to create supervised learning models.

Now let us do more experiments on this data but with multi-layer perceptron.
https://scikit-learn.org/stable/modules/neural_networks_supervised.html

(1) [8 pts] Use the scikit-learn function MLPClassifier to classify the examples into the three species in the dataset. Use the partition function you created in early HWs to split the data into training and validation sets. Create a multi-layer perceptron with 1 hidden layer with 5 hidden nodes (this means 1 hidden layer plus an output layer). Train the model on your training data and test the model on your validation data, and plot the validation model performance in terms of iterations.

(2) [8 pts] Use the same setting you used in (1). Now create a multi-layer perceptron with 2 hidden layers with 3 and 2 hidden nodes respectively. Train the model on your training data and test the model on your validation data, and plot the validation model performance in terms of iterations.

(3) [4 pts] Compare the validation performance of the two models and discuss which one performs better overall (with iterations). You can try different solvers (lbfgs, sgd, or adam).